

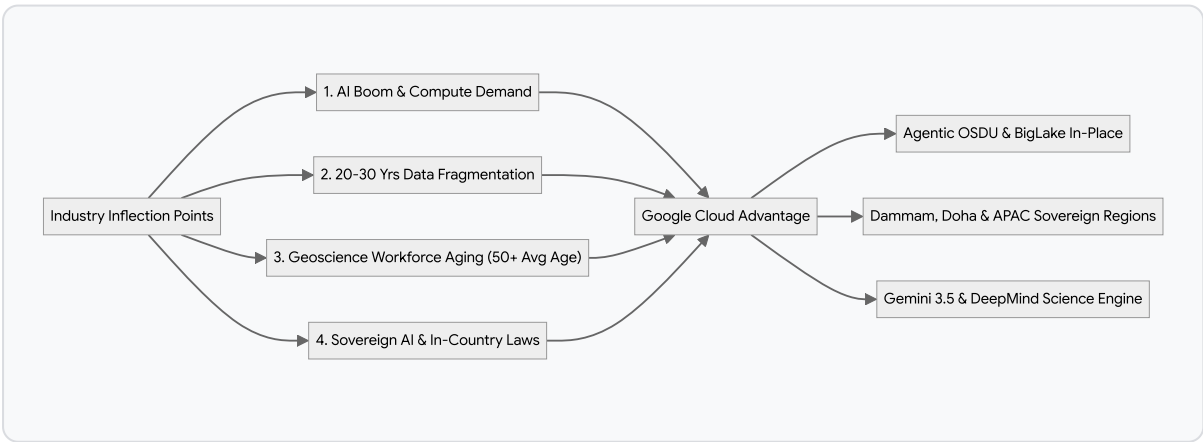
Google Cloud Oil & Gas Industry Strategy & Execution Plan

Document Title: Executive Strategy & Market Execution Roadmap **Target Audience:** Google Cloud Executive Leadership (C-Suite, VP Industry Solutions, Regional VPs) **Author:** Google Cloud Global Oil & Gas Industry Strategy Team **Date:** August 2026 **Classification:** STRATEGIC EXECUTIVE BRIEFING · CONFIDENTIAL

IMPORTANT: Executive Summary & Sizing the Prize Energy operators represent a **\$9.5B Serviceable Addressable Market (SAM)** for Google Cloud by 2028. Driven by the convergence of the **AI inflection point**, an urgent **data fragmentation crisis**, and mandatory **sovereign cloud requirements**, Google Cloud is positioned to expand its market share from **10% today to 20%+ by 2028**, capturing **\$1.96B in annual recurring revenue** across named oil & gas accounts.

1. The Strategic Imperative: Why Oil & Gas Now?

The global oil & gas sector generates **\$4.9 Trillion in annual revenue** (IEA World Energy Outlook) and spends **\$44 Billion annually on IT and digital infrastructure** (Gartner IT Spending Forecast). However, public cloud adoption in oil & gas currently stands at only **~18%** (IDC Cloud Tracker).



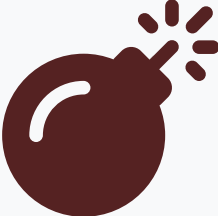
Macro Industry Drivers

1. AI Inflection Point & Compute Demand: Energy operators sit on both sides of the AI boom — as operators requiring AI to optimize field recovery, and as energy suppliers powering the

massive power demand of AI data centers. 2. **Data Fragmentation Crisis:** Operators manage 20–30 years of siloed data across OSDU, PI historians, legacy paper logs, and ERPs. The cost of data fragmentation is now a board-level risk. 3. **Workforce Aging Crisis:** With the average upstream geoscientist age exceeding **50+ years** ([SPE Workforce Survey](#)), agentic AI that encapsulates domain expertise is a vital workforce continuity strategy. 4. **Sovereign AI Mandates:** NOCs in Saudi Arabia, Qatar, Indonesia, Thailand, and Japan mandate in-country processing under strict national data residency laws ([Saudi NCA Regulations](#)). 5. **Energy Transition & CCUS:** Annual CCUS investment surpassed **\$5B+** ([Global CCS Institute](#)), requiring autonomous MRV (Measurement, Reporting, and Verification) software platforms.

2. Market Opportunity & Sizing the Prize

Google Cloud's addressable market in oil & gas is structured across three concentric tiers, representing a total addressable market of **\$21.0B by 2028**:



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Market Sizing Matrix (TAM / SAM / SOM)

Market Segment	2028 Valuation	Scope & Definition	Data Source
Total Addressable Market (TAM)	\$21.0 Billion	Total global cloud, data, and AI infrastructure spending across all upstream, midstream, and downstream O&G operators.	IDC Cloud Tracker / Straits Research
Serviceable Addressable Market (SAM)	\$9.5 Billion	High-margin, Google-winnable workloads: Agentic AI, Sovereign NOC cloud, Subsurface HPC, and CCUS MRV.	Gartner O&G Vertical Analysis
Serviceable Obtainable Market (SOM)	\$3.2 Billion	3-year realistic capture pipeline across 33 named Tier 1–4 accounts and global NOCs.	Google Cloud Internal Pipeline Analysis

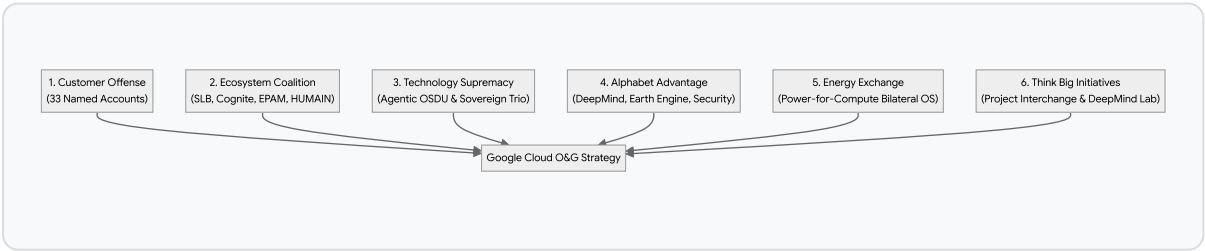
Cloud Market Share Benchmark (O&G Vertical)

Cloud Provider	Current Market Share	Key Anchor Accounts	3-Year Target
AWS	37%	Shell, Oxy, Aker BP, TGS, ENI, Petrobras	32%
Azure	32%	Chevron, Equinor, Devon, ADNOC, Petronas	28%
Google Cloud	10%	Pertamina, PTTEP, Reliance, TotalEnergies (via SLB)	20% (\$1.96B ARR)
Others / On-Prem	21%	Regional NOCs, Private Datacenters, National Clouds	20%

5-Year Revenue Projection by Workload Pillar (\$ Millions)

Workload Category	2024	2025	2026	2027	2028	5-Yr CAGR
AI/ML Workloads	\$80M	\$140M	\$250M	\$450M	\$750M	75%
HPC Seismic & Simulation	\$40M	\$65M	\$110M	\$180M	\$280M	63%
Data Analytics & OSDU	\$120M	\$180M	\$260M	\$380M	\$520M	44%
Sovereign Cloud Regions	\$30M	\$55M	\$95M	\$160M	\$260M	71%
CCUS & Carbon MRV	\$5M	\$15M	\$40M	\$80M	\$150M	134%
TOTAL REVENUE	\$275M	\$455M	\$755M	\$1,250M	\$1,960M	48%

3. The 6-Pillar Operating Strategy



Pillar 1: Customer Offense

Four-tier account attack strategy prioritizing 33 named operators. High-velocity pursuits in Tier 1A (EQT, Devon, Diamondback, Harbour, TGS) combined with sovereign NOC engagement in Tier 2 (Aramco, KOC, Pertamina, PTTEP, Inpex, QatarEnergy).

Pillar 2: Ecosystem Coalition

Deep co-sell integrations with primary energy ISVs (**SLB Delfi/Sequestri, Cognite Data Fusion, Baker Hughes Cordant, AspenTech**), premier GSIs (**EPAM, Accenture, TCS/Infosys**), and sovereign AI partners (**HUMAIN AI** in Saudi Arabia).

Pillar 3: Technology Supremacy

Deploying **Agentic OSDU** and non-OSDU data lakes via BigQuery Omni and BigLake. Providing zero-egress in-place query over incumbent AWS S3 / Azure ADLS repositories paired with TPU v5p/v6 supercomputing clusters for subsurface HPC.

Pillar 4: Alphabet Advantage

Packaging cross-Alphabet innovations into industrial offerings: **DeepMind** (molecular discovery and weather risk), **Google Earth Engine** (methane monitoring & ROW pipeline tracking), **Workspace with Gemini** (workforce transformation), and **Mandiant/Wiz** (OT/ICS cybersecurity).

Pillar 5: Energy Exchange

Structuring bilateral industrial deals: Leveraging Google's massive clean energy procurement to contract firm power from energy operators in exchange for enterprise GCP compute and AI transformation commitments.

Pillar 6: THINK BIG Initiatives

Launching multi-billion-dollar R&D and commercial initiatives: **Project Interchange** (Energy-AI Bilateral OS), **DeepMind Energy Lab** (Frontier materials discovery for hydrogen & CCUS), and **CCUS Transformation Partnerships** with anchor consortia (ExxonMobil LCS, East Coast Cluster, INEOS Greensand).

4. Customer Revenue Map (33 Named Accounts)

NOTE: Figures below reflect estimated total addressable cloud/AI opportunity per account, 3-year revenue ramp projection, and current pipeline deal stage. "Actual GCP Spend" is managed via CRM/Salesforce records.

Account Name	Tier	Region	Incumbent	Strategy Posture	Est. Opp (\$M)	3-Yr Ramp (Y1/Y2/Y3) (\$M)	Pipeline Stage
Saudi Aramco	2	MEA	CNTXT / Azure	Lead / Sovereign	\$150M	[\$20M, \$60M, \$150M]	Qualifie
ExxonMobil	3	Americas	Azure	Multicloud Wedge	\$120M	[\$10M, \$40M, \$80M]	Prospec

Account Name	Tier	Region	Incumbent	Strategy Posture	Est. Opp (\$M)	3-Yr Ramp (Y1/Y2/Y3) (\$M)	Pipeline Stage
TotalEnergies	3	EMEA	Multi	Lead (via SLB)	\$80M	[\$15M, \$40M, \$80M]	Qualifie
QatarEnergy	2	MEA	Greenfield	Sovereign Lead	\$60M	[\$8M, \$25M, \$60M]	Prospec
Reliance Industries	3	APAC	GCP Anchor	Lead / Expansion	\$60M	[\$18M, \$35M, \$60M]	Product
BP	3	EMEA	Multi (Azure/AWS)	Multicloud Wedge	\$55M	[\$5M, \$20M, \$40M]	Prospec
ConocoPhillips	1A	Americas	Azure	Multicloud Wedge	\$50M	[\$5M, \$15M, \$30M]	Prospec
Chevron	Fortress	Americas	Azure	Co-Exist Wedge	\$50M	[\$5M, \$18M, \$35M]	Prospec
Pertamina	2	APAC	GCP Anchor	Sovereign Lead	\$45M	[\$12M, \$28M, \$45M]	PoC
ADNOC	2	MEA	Azure / G42	Co-Exist / Sovereign	\$45M	[\$5M, \$15M, \$30M]	Prospec
Petrobras	3	Americas	AWS	Multicloud Wedge	\$45M	[\$5M, \$15M, \$35M]	Prospec
Devon Energy	1A	Americas	Azure	Lead / Win-Back	\$40M	[\$10M, \$25M, \$40M]	Prospec
KOC / KPC	2	MEA	Greenfield	Sovereign Lead	\$40M	[\$8M, \$20M, \$40M]	Prospec
Shell	Fortress	EMEA	AWS	Co-Exist Wedge	\$40M	[\$5M, \$15M, \$30M]	Qualifie

Account Name	Tier	Region	Incumbent	Strategy Posture	Est. Opp (\$M)	3-Yr Ramp (Y1/Y2/Y3) (\$M)	Pipeline Stage
EQT / Expand Energy	1A	Americas	Greenfield	Lead / Fast-Track	\$35M	[\$8M, \$20M, \$35M]	Qualifie
Petronas	2	APAC	Azure	Multicloud Wedge	\$35M	[\$5M, \$15M, \$25M]	Prospec
ENI	3	EMEA	AWS	Multicloud Wedge	\$35M	[\$5M, \$15M, \$30M]	Prospec
Equinor	Fortress	EMEA	Azure	Co-Exist Wedge	\$35M	[\$3M, \$12M, \$25M]	Prospec
PEMEX	3	Americas	Greenfield	Sovereign Lead	\$35M	[\$5M, \$15M, \$25M]	Prospec
TGS Energy Data	1A	EMEA	AWS	Win-Back Pursuit	\$30M	[\$8M, \$18M, \$30M]	Qualifie
Oxy	Fortress	Americas	AWS	Co-Exist Wedge	\$30M	[\$3M, \$12M, \$25M]	Prospec
ONGC	2	APAC	Greenfield	Sovereign Lead	\$30M	[\$5M, \$15M, \$30M]	Prospec
Diamondback	1A	Americas	Greenfield	Lead	\$25M	[\$5M, \$15M, \$25M]	Prospec
Quantum Capital Group	1B	Americas	Greenfield	PE Sponsor Platform	\$25M	[\$8M, \$18M, \$25M]	Qualifie
PTTEP	2	APAC	GCP Anchor	Sovereign Production	\$25M	[\$8M, \$16M, \$25M]	Product
Repsol	3	EMEA	Azure	Multicloud Wedge	\$25M	[\$3M, \$10M, \$20M]	Prospec

Account Name	Tier	Region	Incumbent	Strategy Posture	Est. Opp (\$M)	3-Yr Ramp (Y1/Y2/Y3) (\$M)	Pipeline Stage
Woodside Energy	3	APAC	AWS	Multicloud Wedge	\$25M	[\$3M, \$10M, \$20M]	Prospect
Hess Corporation	3	Americas	AWS	Multicloud Wedge	\$25M	[\$3M, \$10M, \$20M]	Prospect
Harbour Energy	1A	EMEA	Greenfield	Lead	\$20M	[\$5M, \$12M, \$20M]	Prospect
EOG Resources	1A	Americas	AWS	Multicloud Wedge	\$20M	[\$3M, \$10M, \$20M]	Prospect
Continental Resources	1C	Americas	Greenfield	Lead	\$20M	[\$5M, \$12M, \$20M]	Prospect
Inpex	2	APAC	Greenfield	Sovereign Lead	\$20M	[\$3M, \$10M, \$20M]	Prospect
Santos	3	APAC	AWS	Multicloud Wedge	\$20M	[\$3M, \$8M, \$18M]	Prospect
YPF	3	Americas	Greenfield	Lead	\$20M	[\$3M, \$10M, \$18M]	Prospect
Williams Companies	4	Americas	Azure	Midstream Wedge	\$18M	[\$3M, \$8M, \$15M]	Qualified
Aker BP	1A	EMEA	AWS	Multicloud Wedge	\$18M	[\$3M, \$8M, \$18M]	Qualified
EnCap Investments	1B	Americas	Greenfield	PE Sponsor Platform	\$15M	[\$3M, \$8M, \$15M]	Prospect
Mewbourne Oil	1C	Americas	Greenfield	Lead	\$8M	[\$2M, \$5M, \$8M]	Prospect

5. Multicloud Wedge & Sovereign Cloud Architecture



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The Co-Existence Wedge Doctrine

1. **Zero Migration Requirement:** Leave incumbent S3/ADLS repositories untouched. Position BigQuery Omni and BigLake for in-place federated queries over AWS/Azure stores. 2. **Gemini Agentic Reasoning:** Execute long-context reasoning over unstructured LAS/DLIS well logs, seismic surveys, and production reports without moving raw petabytes. 3. **Subsurface HPC Bursting:** Provide TPU v5p/v6 supercomputing clusters for reservoir simulation and reverse time migration (RTM) bursting at 40% lower TCO than AWS EC2.

Sovereign Region Footprint Matrix

Sovereign Region	Target Accounts	Governance & Control Model
Dammam (`me-central2`)	Saudi Aramco, CNTXT	Class C CST/NCA certified; CNTXT-operated External Key Management with Key Access Justifications.
Doha Region	QatarEnergy, LNG Ecosystem	In-country Qatar data boundary; low-latency regional AI inference for LNG shipping & trading.
Jakarta (`asia-southeast2`)	Pertamina	In-country Indonesia data residency; aligned with Pertamina Digital Hub IT Strategic Plan 2025–2029.
Bangkok (`asia-southeast1`)	PTTEP	In-country Thailand data residency; Apigee/BigQuery/GKE architecture for Net Zero analytics.
Tokyo / Osaka	Inpex	In-country Japan data residency; dual-region resilience for Ichthys LNG & CCUS monitoring.
GDC Air-Gapped	Kuwait NOCs, ADNOC OT	Disconnected Google Distributed Cloud hardware deployed directly on-premise at critical OT sites.

6. Competitive Intelligence & Capability Matrix



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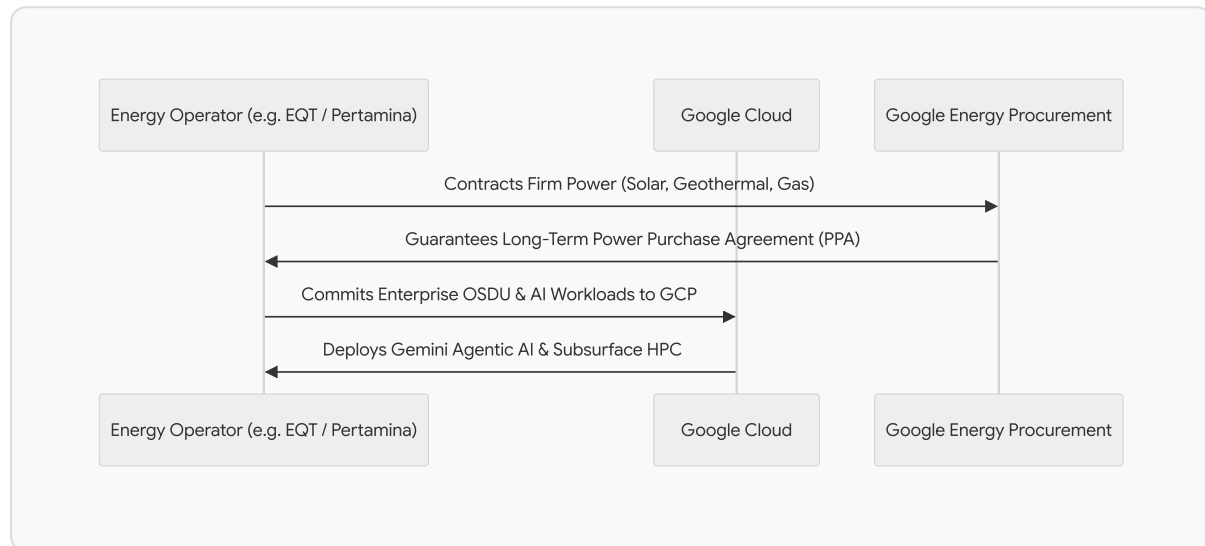
Head-to-Head Capability Matrix

Strategic Capability	Google Cloud	AWS	Microsoft Azure	Google Competitive Advantage
Agentic AI & Reasoning	Gemini 3.5 / 3.6 (1M+ Token)	Bedrock / Claude 3.5	Azure OpenAI / GPT-4o	Long-context multimodal reasoning over full seismic & well log repositories.
Multicloud In-Place Query	BigQuery Omni & BigLake	Redshift Spectrum	Synapse Link	Zero-egress query execution over incumbent AWS S3 / Azure ADLS lakes.
Subsurface HPC Compute	TPU v5p / v6 Supercomputers	EC2 UltraClusters	H100/H200 VMs	40% lower TCO for massive seismic inversion & reservoir simulation sweeps.
Sovereign Cloud Controls	KSA (Dammam), Doha, GDC Airgap	AWS Top Secret	Azure Sovereign	Direct partnership with CNTXT (KSA) and disconnected GDC Air-Gapped hardware.
Energy Procurement OS	Project Interchange	Clean Energy Contracts	PPA Deals	Bilateral OS: Swapping data center power demand for enterprise cloud commitments.
CCUS & Carbon Accounting	SLB Sequestri + Earth Engine	AWS Carbon	Azure Sustainability	Integrated satellite methane monitoring (MethaneSAT) + SLB reservoir modeling.

7. Strategic "Think Big" Initiatives

Project Interchange: Energy-AI Bilateral Operating System

Project Interchange establishes a bilateral commercial framework with energy majors. Google Cloud contracts firm clean power generation (solar, geothermal, nuclear) from operators (e.g., EQT, Pertamina, TotalEnergies) to supply Google's expanding AI data center estate. In return, the operator commits their enterprise IT, OSDU data lakes, and AI workloads to Google Cloud.



DeepMind Energy Lab: Frontier Science Engine

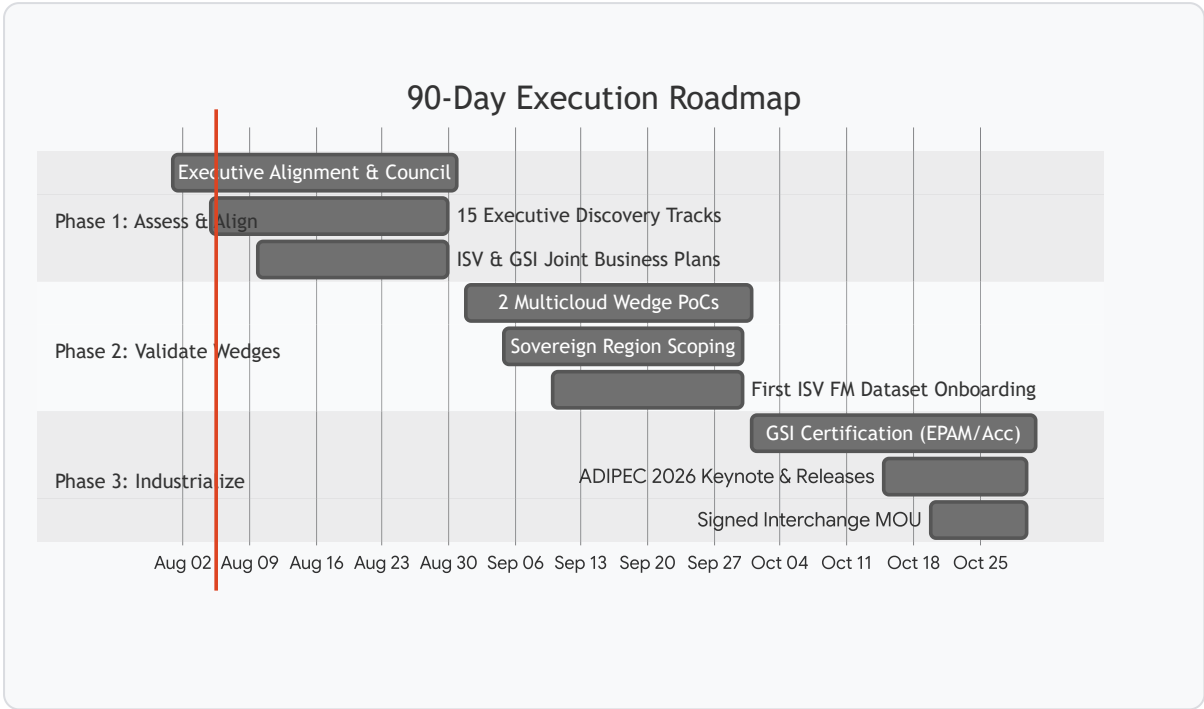
The DeepMind Energy Lab accelerates frontier R&D for energy transformation:

- **Molecular Discovery (GNoME & AlphaFold 3):** Computational screening of 100,000+ candidate solvent molecules for CCUS carbon capture and novel corrosion-resistant alloys for H₂ pipeline blending.
- **Physics Simulation (TORAX Architecture):** Differentiable neural proxy simulators executing full-physics reservoir simulations **10,000× faster** than legacy solvers.
- **Autonomous Operations:** Reinforcement learning policies for real-time autonomous drilling control and process safety precursor detection.

CCUS Transformation Partnerships

In partnership with **SLB (Delfi/Sequestri)**, Google Cloud establishes strategic technology partnerships with anchor CCUS consortiums: 1. **ExxonMobil Low Carbon Solutions:** Intelligent Gulf Coast Hub platform for multi-tenant carbon accounting and EPA Class VI compliance. 2. **East Coast Cluster (BP / Equinor / TotalEnergies):** Neutral multi-operator CCUS intelligence platform. 3. **INEOS Greensand (Denmark):** Enterprise cloud expansion leveraging existing SLB Delfi deployment on GCP.

8. Execution Roadmap & Leadership Alignment



30-60-90 Day Execution Workstreams

Execution Phase	Focus Theme	Key Exit Deliverables
Days 1–30	Assess, Align & Activate	• Global O&G Deal & Product Council operationalized. • Top 33 accounts scored on priority heat map. • 15 customer discovery tracks opened (Pertamina, PTTEP, EQT, Devon). • Project Interchange framework ratified with Google Energy Procurement.
Days 31–60	Validate Wedges & Sovereignty	• 2 multicloud wedge validations live (1 AWS, 1 Azure). • Sovereign AI roadmaps signed for Aramco (Dammam) and Pertamina (Jakarta). • First ISV dataset onboarded to Vertex AI for Foundation Model pre-training.
Days 61–90	Scale, Accelerate & Penetrate	• 3 solution kits shipped; EPAM & Accenture certified on Agentic OSDU. • Marquee customer announcements locked for ADIPEC 2026. • 1 Project Interchange MOU signed; 1 DeepMind Energy Lab partnership signed.

Strategic Objectives (OKRs)

- **OKR 1:** Sign **3–5 Tier 1 accounts** (EQT, Devon, Diamondback) with ≥1 live in production on Agentic OSDU.
- **OKR 2:** Validate **2 multicloud wedges** in AWS/Azure fortress accounts (Shell, Chevron) with zero migration escalations.
- **OKR 3:** Ratify **1 Gulf NOC sovereign PoC** (Aramco/KOC) and launch HUMAIN AI enablement on Dammam `me-central2`.
- **OKR 4:** Achieve ≥**30% partner-originated pipeline** via SLB, Cognite, Baker Hughes, EPAM, and Accenture.
- **OKR 5:** Embed **3 Alphabet modules** (Earth Engine, WeatherNext, Mandiant) across active account pursuits.
- **OKR 6:** Execute **1 signed Project Interchange MOU** and **1 DeepMind Energy Lab research partnership** at ADIPEC 2026.

Executive Decisions Required from Google Leadership

CAUTION: Executive Alignment & Resource Requests 1. **Resource Concentration:** Ratify the 33-account focus model, assigning direct P1 executive sponsors. 2. **Engineering Staffing:** Formally assign Google product/engineering leads to the 6 Agentic OSDU reference patterns. 3. **Sovereign Region Investment:** Accelerate feature parity for Dammam (`me-central2`) and Doha regions. 4. **Project Interchange Approval:** Authorize Google Energy Procurement to structure bilateral power-for-compute deals with EQT and Pertamina.

Footnotes & Data Sources

- [IEA World Energy Outlook 2025](#) — Global Energy Demand & Revenue Benchmarks
- [Gartner IT Spending Forecast Q2 2025](#) — Vertical IT Expenditure Metrics
- [IDC Cloud Tracker 2025](#) — Enterprise Public Cloud Adoption Rates
- [Precedence Research](#) — Artificial Intelligence in Oil & Gas Market Projection (\$5.4B by 2035)
- [MarketsandMarkets](#) — Digital Oilfield Market Forecast (\$43.0B by 2029)
- [Global CCS Institute Status Report](#) — Global CCUS Project & Investment Tracker
- [SPE Journal of Petroleum Technology](#) — Upstream Geoscience Workforce & Digital Energy Surveys